

# ADVANTAGE Section Discussion



Drilling Fluids

|                                                |                            |                                            |                              |
|------------------------------------------------|----------------------------|--------------------------------------------|------------------------------|
| <b>Operator</b> Peak Petroleum Indust. Limited | <b>Field</b> Bilabri       | <b>Block/Co/Parish</b> OML - 122 / Nigeria | <b>Facility</b> Bilabri DX 1 |
| <b>Well</b> Bilabri DX 1                       | <b>Rig</b> BULFORD DOLPHIN | <b>Job</b> Bilabri DX 1                    |                              |

## Section Details - 6" Section - #6

|                                                                                         |                                  |                         |                             |
|-----------------------------------------------------------------------------------------|----------------------------------|-------------------------|-----------------------------|
| <b>Mud Type</b> Invert Emulsion                                                         | <b>Mud System</b> SYN-TEQ (PARA) | <b>Top MD</b> 3200.00 m | <b>Bot MD</b> 2689.00 m     |
| <b>Start-End Date</b> 26/Jan/2006-26/Feb/2006                                           | <b>Open Hole Diam</b> 8 1/2 in   | <b>Total Days</b> 31    | <b>Sect Length</b> 517.00 m |
| <b>Section Comment</b> Not plan to drill 6" Unconclude 8.5 section due to sever losses. |                                  |                         |                             |

## Section Discussion

Make up new 6" slick BHA. Trip in and tag cement at 3011m. Drilled out cement to 3059m, circulated to condition mud 12.9 ppg, pumped slug and pull out for CBL Log. Rig up & run Baker Atlas' CBL log on wireline.

Trip back in hole with 6" BHA to tag cmt at 3059m. Circulate, and commenced drilling cement and shoe to 30362m. Circulated bottoms up with 13 ppg mud and performed Shoe Bond test to EMD= 15.5 ppg. Circulate and raise mud weight to 13.5 ppg. Wash and ream 8.5" rate hole from 3062m to 3110m. Observed gas increase to 1810 units. Increase mud weight to 13.7 ppg & continued to wash/ream from 3110 to 3200m.

While increasing mud weight to 13.9 ppg circulated bottoms up and continue to circulate and raise mud weight to 14.1 ppg. Spot 65 bbls of sweep on bottom and perform wiper trip to shoe. O.K Return to bottom, circulated hi-vis pill out and raise mud weight once more to 14.3 ppg. Pull out 5 stands and slug pipe. Pull to surface for new BHA.

Make new 6" BHA and run in. Ream/wash last stand and drilled ahead from 3200 to 3227m. Observed power failure. Pull to shoe and raise mud weight to 14.5pp.

Continue to investigate on electrical problem. Unable to drill due to electrical problem. Wait for expert. Circ hole while at the shoe for 20 mins every 2 hrs.

Trip in hole after a couple of days at the shoe with no fill or drag was observed. Ream last stand as a precaution. Circulated bottoms up while conditioning mud to 14.5 ppg all round. Pump out of hole & continue to wait on electrical repairs.

With electrical problems fixed, RIH from 7" shoe t/3172m.. Wash/ream t/3200 & then to 6" TD of 3227m. Drill 6" hole t/3234m. Pull back to 8-1/2" hole. Circ & condition mud to 14.5 ppg in/out. RIH t/3230 & wash/ream to 3234m. Continue drilling 6" hole while pumping 30 bbls hi-vis pill every 30m & LCM as required. The mud was treated with base oil to gradually increase the OWR to 80/20 as per orders while maintaining wt & other properties.

Drill t/3545m, circ btms up & work pipe. Pump out of hole f/3545 t/3118 & cont POOH t/3033m. F/check 15 mins inside 7" csg, pump 20 bbls slug. Cont POOH f/3033m. At 2947m, pipe pulled wet. No gain in trip tank. POOH t/2890. Observe well flowing. Space out, shut well in. Stabilize press, SIDP - 440 psi, SICP - 400psi. Attempt to circ out influx by "Driller's Method". Observe increase in DP press f/1050 t/1650 psi followed by rapid increase in csg press. T/tank overflowing. Shut in well on choke, SIDP - 740 psi, , circ riser content. Mud wt 14.6 ppg. Observe shut in press decreasing. Commence " Lubricate & Bleed". Csg press decrease t/2780 psi. No further decrease in csg press. Observe increase in csg press t/2880 psi. Csg press being affected by pumping of 1 bbl vol. Abandon operation. Bring pump on line w/csg press constant. Circ out influx using "Drillers Method" @ 20 SPM/540 psi. Experience frequent periods of no returns.

Wt of returns ranged from 12.9 to 14.6 ppg. Mud thinned down & cut back by formation fluid. Continuously running the degasser. Took returns in separate pit, condition & wt up to 14.6ppg & transfer back to active. Increased active system from 14.6 to 15.0 ppg kill mud & circ @ 40 SPM/950 psi. Cont taking returns of 14.6 ppg to separate pit, wt up to 15 ppg & transfer back into active while swapping pits as the Drillers method of well control continued. More SBM of 11.2 ppg was received from boat, & subsequently weighted up to 15.0 ppg & moved to the active due to the loses.

Close choke. SIDP - 210 psi, SICP - 90 psi. Cont well kill second circ w/Drillers method. Monitor well on choke while wt up mud, SIDPP - 110, SICP - 40. Displace riser t/15.0 ppg mud through chokes. F/C through chokes, F/C through TT, OK. RIH on 5" DP f/2890 t/3175m. No drag. Wash t/3200m. Circ & condition mud @ 65 SPM, 1750 psi. Observe increase in flow. Space out, close upper annula, open upper chokes. SIDPP- 0, SICP - 560 psi. Observe mud loss in riser. Fill same f/TT - 66 bbls. Circ out through chokes w/15 ppg mud @ 20 SPM, 3047 stks. Shut in well on chokes. Observe stabilized SIDPP - 110, SICP - 160 psi. Continue to circ well w/15 ppg mud @ 20 SPM, 3173 stks.

With the continous wting up of cut back mud from well & the prevalent loses, ran out of barite . Close well in & minitor SIDPP - 150 & SICP - 40 psi. Conditioned reserve mud in readiness for wting up while wait on barite from town. 86MT of barite was received & the reserve pits immediately wted up to 15 ppg & moved same to active to continue well control operation. Wt in/out was continously monitored & maintained at 15 ppg. The system was gradually treated with LCM (Solulflake fine & Calcium Carbonate F/M) to cure seepage losses. Additives for rheology, oil wetting, filtration loss ES was added to mud while circulating.

While circ well on choke, DP readings were eratic. Shut in well, SIDPP - 150, SICP - 350 psi. Cont circ @ 20 SPM, 580 psi. Left Auto choke plugging. Shut in, SIDPP - 160, SICP - 510 psi. Cont circ with right Auto choke @ 20 SPM. Circ btms up with final 5000 stks pumped on wide open choke with csg press 0 psi. Losing 24 bbls/hr. Shut down pump. F/C through choke into TT. Open well, observe 4.5 loss to hole. Fill up riser. Monitor well - static. Circ conventionally @ 30 SPM, 525 psi.

Gradual addition of CaCO3 F/M in the active system was maintained in the active system to cure seepage losses while circulating in open hole. Conditioned mud & continued monitoring mud weight in and out and maintaining same at 15.0 ppg. More mud was received from boat to compensate for the loses.

Hole was circulated clean @ 40 SPM. Attempted to go down but no success. Check ram on BOP OK. Circulated bottoms up, F/C, pump out of hole f/3195 t/2711m, top of 7" liner. F/C, Circulate bottoms up, F/C OK. Continue to POOH, monitoring volumes f/2711 t/2434m (wet tripping)

While tripping, more barite & 893 bbls of SBM of average wt of 14.0 ppg mud was received & 241 bbls of 11.5 backloaded. The received mud was conditioned & wtcd up to 15.0 ppg.

The trip out of hole continued t/2434 & gain was observed on F/C. CBU @ 30 SPM. Stop short of 200 stks to btms up due to observed pit gain. Shut in well, SIDPP - 480, SICP - 380 psi. Circ @ 20 SPM. Shut on choke, SIDPP - 1140, SICP - 1830 psi. Close upper pipe rams, space out & commence bullheading. Pump #1 - 11 SPM down string, Pump #3 - 11 SPM down choke. Total strks 7564, 620 bbls. Shut down pumps & monitor press. SIDPP - 600, SICP - 890 psi. Bullhead a further 250 bbls of 15.0 ppg mud. Pump #1 - 12 SPM down string, Pump #3 - 24 SPM down choke. Stop pumps, shut in with SIDPP - 380, SICP - 380 psi. & monitor well & wait on barite.

With the bullheading of 15 ppg mud down hole to kill well, additional 100MT barite & 524 bbls SBM (average wt = 14.0 ppg) was received, conditioned & wtcd up to 15.0 ppg.

Bullhead 250 bbls 15.0 ppg mud w/mud pump #1 on string - 12SPM & M/pump #3 on choke - 24SPM. Close insert PKR. Reduce annular press to 600psi, bleed off trapped gas b/w annular & pumps, none. Open insert PKR, install TIW in string. Cont t/monitor well. Bullhead 150 bbls 15 ppg mud into formation. Press f/700 t/580 psi @ 40 SPM. Shut down pumps, open choke, observe well. Press 0.8 bbls increase in tnk over 15 mins, then OK. RIH f/1559 t/2940, F/C @ 2291m. M/U TDS, insert side entry & safety valve.

With a total of 1,270 bbls bullheaded down hole within this period, the hole seemed stable for a cmt job to plug off the open hole from where the influx was coming from.

The initial cmt job couldn't hold. Had to patch hole in mix water transfer line t/cmt unit & change blocked valve. Attempt to mix cmt, no success due to foaming in mix water. Shut down cmt job, dump & clean slug pit for mix water while wait on chem f/town. CBU through choke @ 50 SPM, initial press 1600 psi, reducing t/800 psi. Reduce stk to 40 SPM. Circ riser content - 80SPM. Open well & observe - OK. Close annular & perform injection rate test. Test lines, perform cmt job & displace cmt f/string & t/60m below bit @ 3000m. 117 bbls of cmt slurry + 20 bbls water ahead & 10 bbls water behind was pumped. Bleed off press @ BJ unit & record flow back volume of 3 bbls. POOH f/2940 t/2850m (wet). Circ conventionally drill string volume & 35 SPM. Close upper annular & apply 500 psi on annulus & wait on cmt.

Water content in mud increased after cmt job & circulation. Maintained wt at 15.0 ppg with barite & base oil & treating mud with base oil, emulsifiers, rheology modifiers etc, while circulating.

The cmt plug was pressure tested to 1000 psi, 30 mins - OK. Circ btms up @ 62 SPM. RIH to 3100m, no cmt. Circ @ 120 strks. Pull back t/2940m. Circ btms up @ 140 strks. F/C 10 mins - OK. Pump 20 bbls slug, POOH f/2805 t/498m. It was discovered that 498.75m of BHA was missing down hole.

A fishing BHA was RIH t/2666m. Trip out did not show any fish.

A 4-3/4" BHA was M/U & RIH t/2500m. Circ btms up @ 100 SPM. At btms up split manifold & boost riser @ 80 SPM & 80 SPM on hole. Wash/ream f/2500 t/2660m. Drill cmt stringers f/2660 t/2662m. Drill cmt f/2662 t/2695m. Circ btms up, F/C, pump slug, POOH. Continued to monitor & maintain mud weight in/out at 15.0 ppg while washing down to top of cmt & while drilling out cement. Active system was treated with emulsifiers and base oil to take care of water from drilling out cement.

The csg was later tested to 1000 psi for 30 mins & thereafter a test was performed on the BOPs.

An 8-1/2" assy was made & RIH. While tripping, 2 bad shaker screens on shakers 1 and 2 were replaced with 2 x 210 new screens and 2 bad screens on shakers 2 and 3 with 2 x 175 used screens. The screens were changed back to 3x150s on shakers 1&2 and 3x125s on shaker 3 to accommodate flow.

The csg was washed & reamed f/2500 t/2695m. While maintaining mud wt @ 15.0 ppg in/out & conditioning mud, attempt to drill cmt was not successful. A trip was made out of hole & ran back in with a 9-5/8" bit & Scraper. Tag cmt @ 2695m, Pump 40 bbls hi-vis, hi-wt pill & circ out of hole. Pump slug, POOH.

A Bridge Plug was set on wireline @ 2689m. Set plug at 2689m. A CBL was subsequently performed on w/L. The bridge plug was pressure tested against shear rams at 3000 psi for 30 mins.

Laying down of 4 3/4" DC and 3 1/2" DP commenced while pit cleaning for brine take on was in progress.